

CHAROTARUNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Pharm. Examination (New)

University Theory Examination December 2015

PH112 – Human Anatomy and Physiology - I

Date: 21/12/2015, Monday

Time: 10.00 a.m. to 1.00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION - I

Q.1 Attempt all questions. Each question is of one mark.

20

1. Most of the glucose that is filtered through the glomerulus undergoes reabsorption in the
 - A. descending limb of the loop of Henle
 - B. ascending limb of the loop of Henle
 - C. distal tubule
 - D. proximal tubule
2. In RBC lifecycle Iron is removed from heme portion in form of Fe^{3+} through which transporter?
 - A. Erythropoietin
 - B. Transferrin
 - C. Ferritin
 - D. Globin
3. Which type of 'WBC cells' Nucleus have a shape of Kidney?
 - A. Monocyte
 - B. Basophils
 - C. Neutrophil
 - D. Lymphocyte
4. Which of the following is not a part of plasma protein?
 - A. Albumin
 - B. Fibrinogen
 - C. Bilirubin
 - D. Globulin
5. Which of the following considered as Natural Killer cells in immune response?
 - A. Lymphocyte
 - B. Monocyte
 - C. Eosinophils
 - D. Basophils

6. A basic mechanism involved in concentrating the urine is
A. Diffusion
B. Countercurrent multiplier system
C. Active transport
D. Passive transport
7. Choose the correct sequence of current flow through the heart wall.
A. AV node, SA node, Purkinje fibers, AV bundle of His, right and left bundle branches
B. SA node, AV node, AV bundle of His, right and left bundle branches, Purkinje fibers
C. SA node, Purkinje fibers, AV node, AV bundle of His, right and left bundle branches
D. AV node, Purkinje fibers, AV node, AV bundle of His, right and left bundle branches
8. During pulmonary circulation blood leaves the
A. right ventricle and goes directly to the aorta.
B. right atrium and goes directly to the lungs.
C. right ventricle and moves to the lungs.
D. right atrium and goes directly to the left ventricle.
9. Pyelonephritis is
A. dislocation of kidney
B. tumour of kidney
C. inflammation of kidney
D. infection of kidney
10. The walls of the alveoli are composed of two types of cells, type I and type II. The function of type II is to _____
A. trap dust and other debris
B. secrete surfactant
C. replace mucus in the alveoli
D. protect the lungs from bacterial invasion
11. The region where a bronchus and pulmonary vessel enter and exit the lung is
A. the capsule
B. the base
C. the hilum
D. the apex
12. Which of the following occurs during the chloride shift?
A. Bicarbonate is removed from the blood
B. Chloride is removed from the erythrocyte
C. Bicarbonate is removed from the erythrocyte
D. Chloride is exchanged for bicarbonate
13. Which one of this is the 12th factor in clotting process?
A. Fibrin stabilizing factor
B. Fibrinogen
C. Hageman factor
D. Christmas factor
14. Which chemical is a strong inhibitor of Platelet adhesion & release?
A. Heparin
B. Prostacyclin
C. TXA2
D. Plasminogen

15. All of the following are accessory digestive organs except _____.
 - A. the teeth
 - B. salivary glands
 - C. the liver
 - D. the spleen
16. What is the full form of RAAS?
 - A. Renin angiotensin aldosterone system
 - B. Renin adrenergic system
 - C. Renin alkaline aldosterone system
 - D. Reticulo angiotensin aldosterone system
17. Which of the following is not a function of the stomach?
 - A. formation of chime
 - B. secretion of gastric juice
 - C. absorption of fats
 - D. storage of ingested food
18. One of the following is not a disease / condition of Respiratory system.
 - A. Asthma
 - B. Bronchitis
 - C. Cystitis
 - D. COPD
19. The exocrine secretory units of the pancreas are called _____.
 - A. Beta cells
 - B. Islet cells
 - C. Acini
 - D. Alpha cells
20. The function of G-cells is to _____.
 - A. secrete gastrin hormone
 - B. secrete gastric acid
 - C. secrete gastric lipase
 - D. secrete somatostatin

SECTION – II

Q 2 Attempt any **FOUR** of the following

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|---|---|----|
| A | Explain a brief physiology of cardiac contraction & enlist major disorders of cardiovascular system | 05 |
| B | Explain in brief about blood circulation in cardiovascular system | 05 |
| C | Provide detail anatomy of the heart with diagram | 05 |
| D | Explain concept of cardiac cycle in detail | 05 |
| E | Explain regulation of Blood pressure in body and factors affecting the same | 05 |
| F | Give following definitions:
ECG, Heart rate, Blood pressure, Cardiac out put, heart sound | 05 |

SECTION – III

Q 3 Attempt any FOUR of the following

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|----|---|---|
| 05 | A | Describe the location, anatomy and functions of the stomach. |
| 05 | B | Define hemostasis. Discuss the blood clotting mechanism. |
| 05 | C | Draw a neat labeled diagram of nephron and explain urine formation. |
| 05 | D | Explain physiology of respiration. |
| 05 | E | Define the following terms. |
| | | 1. Tidal volume |
| | | 2. Vital capacity |
| | | 3. Anatomy and physiology |
| | | 4. Renal clearance |
| | | 5. Deglutition |
| 05 | F | Discuss the blood groups and blood transfusion. |

Q 4 Attempt any FOUR of the following

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|----|---|--|
| 05 | A | What are the components of blood? Write the function of blood. |
| 05 | B | Draw and label a diagram of a liver lobule. Explain the various function of liver. |
| 05 | C | Write the composition and functions of saliva and pancreatic juice. |
| 05 | D | Write a short note on chloride shift. |
| 05 | E | Describe the structure, functions, life cycle, and production of red blood cells. |
| 05 | F | Describe the routes and mechanisms of tubular reabsorption and secretion. |